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Software note

NicheMapR – an R package for biophysical modelling: the ectotherm and Dynamic Energy Budget models

Michael R. Kearney and Warren P. Porter

M. R. Kearney (https://orcid.org/0000-0002-3349-8744) ✉ (mrke@unimelb.edu.au), School of BioSciences, The Univ. of Melbourne, Parkville, VIC, Australia. – W. P. Porter, Dept of Zoology, The Univ. of Wisconsin, Madison, WI, USA.

Ecography
43: 85–96, 2020
doi: 10.1111/ecog.04680

Subject Editor: Brody Sandel
Editor-in-Chief: Miguel Araújo
Accepted 10 September 2019

Mechanistic niche models characterise the fundamental niche of an organism by determining thermodynamic constraints on its heat, water and nutritional budget, and the consequences of this for growth, development and reproduction. They can thus quantify constraints on survival, activity and, ultimately, the vital rates that determine population growth, given a sequence of environmental conditions and the key morphological, physiological and behavioural functional traits.

Here we introduce and document the ectotherm model of NicheMapR, an R package that includes a suite of programs for the mechanistic modelling of heat, water, energy and mass exchange between any kind of ectothermic organism and its environment. The NicheMapR ectotherm model is based on a Fortran program originally developed by Porter, Mitchell and Beckman for predicting core body temperature and evaporative water loss as a function of microclimatic conditions and behavioural thermoregulation. The model includes routines for computing steady state body temperature and evaporative water loss given two extreme microclimates (minimum and maximum shade) as computed by the NicheMapR microclimate model. Behavioural options include posture and colour change, shade-seeking, panting, climbing and retreating underground.

Here we configure the program to be called from R as part of the NicheMapR package and describe the model in detail including new functionality for modelling whole life-cycle energy and water budgets using Dynamic Energy Budget theory. We include scripts for core operation of the ectotherm model as well as stand-alone R scripts for running the DEB model. Example applications are provided in the paper and in the associated vignettes. The integrated microclimate and ectotherm models should provide a strong thermodynamic basis for determining the effects of environmental change on the behaviour, distribution and abundance of ectothermic organisms.

Keywords: biophysical ecology, dynamic energy budget theory, ectotherm, mechanistic niche model, thermoregulation, water budget

The concept of a mechanistic niche model

Organisms can be conceived as open thermodynamic systems (von Bertalanffy 1950), with inherent behaviours and internal processes aimed at maintaining a degree of

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homeostasis and the accumulation of sufficient resources to grow, develop and reproduce. A thermodynamic system is defined by the 'system boundary' which, in the case of an organism, is its outer surface including any 'imaginary' surfaces across open parts of the body, e.g. nostrils and mouth. Across this boundary heat, water and various other metabolic inputs and outputs flow. According to the first law of thermodynamics, these inflows and outflows must balance and either sum to zero or result in storage of the relevant quantity (e.g. energy, mass, momentum) in the organism (Porter 2016). As long as homeostasis is achieved within the tolerable limits, an organism may be able to complete its lifecycle within a given habitat.

Any sequence of environmental conditions that allows for homeostasis to be maintained long enough for the completion of the life cycle can be considered to be within the fundamental niche from a thermodynamic perspective – the 'thermodynamic niche' (Kearney et al. 2013). This spatio-temporal perspective of the niche can account for plastic and ontogenetic changes of the organism as it grows and develops, as well as the consequences of these changes for the environments experienced, in an interactive manner (Levins and Lewontin 1985, Lewontin 2000). Thus, the constraints computed by a mechanistic niche model at the level of the individual provide the life history and phenological information needed to determine vital rates, creating a link to phenomena at the population and distribution levels (Kearney and Porter 2009) (Fig. 1). The niche as defined in this way can be constrained by biotic interactions to a more restricted set of permissible conditions, i.e. realised niche. Moreover, the organism's energy and mass flows in (feeding) and out (breathing, defecation, evaporation and excretion) feed back to the community dynamics (Kooijman et al. 2004).

The thermodynamic niche can be characterised by a set of coupled equations representing the heat budget, the water budget, the nutritional budget and the 'mineral' budget (Fig. 2), as summarised in Kearney et al. (2013), Porter and Gates (1969), Porter et al. (1973), Gates (1980), Porter (1989). The heat budget comprises inputs from metabolism (the aggregate chemical transformations of the organism) and solar (direct and diffuse) and longwave (e.g. emitted from sky and ground) radiation. Heat is also lost by longwave radiation emitted from the animal surface and by evaporation from respiration and bodily surfaces (condensation may add heat) and may be lost or gained by the processes of convection (via wind speed, humidity and air temperature) and conduction to substrates. The water budget and the heat budget interact through the evaporation term, with additional water being lost from excretion and defecation, and being gained from drinking, food and metabolism. The nutritional budget couples to the heat and water budgets through the metabolism and metabolic water terms and captures the uptake of food and its allocation as well as dictating the flows of CO_2, O_2, metabolic water and nitrogenous waste. The equations representing these thermodynamic exchange processes define key morphological, physiological and behavioural traits and

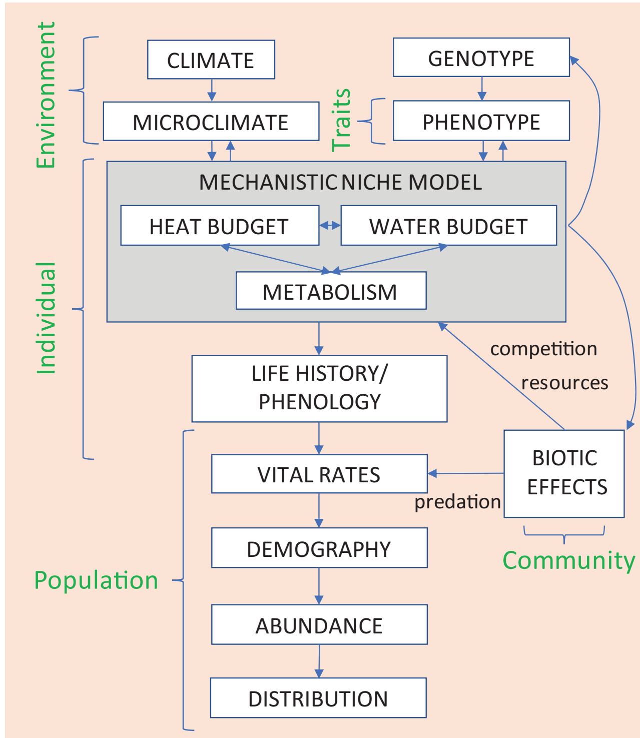

Figure 1. The mechanistic niche modelling scheme in the context of microclimate forcing at the individual level, the interaction between genotype, phenotype and environment and the connection with population and community processes. The coupled heat, water and metabolic processes of the organism as a thermodynamic system (Fig. 2) are individual-level processes that are a function of the organism's traits and the environments it experiences. The organism's state may alter the phenotype or genotype as, for example, through acclimation. The trajectory of growth, development and reproduction resulting from these individual level processes define the life history and phenological patterns that, in turn, affect population processes through the vital rates (fertility, mortality) and ultimately affect distribution and abundance. The thermodynamic constraints are further influenced by biotic interactions between species (competition, predation, disease, food resources) and within species (competition, social interactions).

functional environmental conditions (microclimates) that must be measured or computed. The NicheMapR package provides tools to solve these equations given the available macroclimatic conditions and functional traits (Kearney et al. 2018). It therefore maps the thermodynamic niche to physical conditions available in space and time in a given habitat (Kearney and Porter 2004). It can be used to assess the suitability of a particular location for a species, population or individual, given the habitat and climate. It thereby permits inferences of hard limits on activity, dispersal, distribution and abundance as well as providing quantifications of growth and reproduction potential for use in population and life history analyses.

The microclimate and ectotherm models of the NicheMapR package are now fully open source at

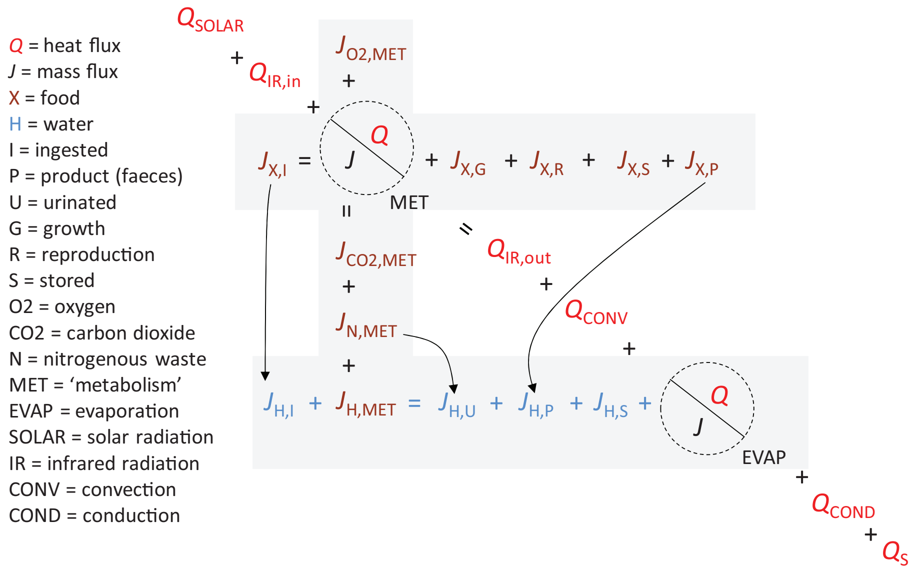

Figure 2. The 'thermodynamic niche' modelling scheme, after Porter and Tracy (1983) and Kearney et al. (2013), showing the coupled heat and mass balance equations that define the organism as an open thermodynamic system. The heat balance terms are in red, the metabolic mass budget is in brown (nutritional and respiratory/excretion budgets), the water budget is in blue and coupling between the energy and mass budget is represented by the dashed circles. The Dynamic Energy Budget Theory provides a mechanistic link between the heat transfer processes and the energy and mass budget.

<https://github.com/mrke/NicheMapR>. The NicheMapR microclimate modelling routines were described in Kearney and Porter (2017). We here provide an overview of the ectotherm subroutines as well as the integration of Dynamic Energy Budget theory into the package. In addition to the information provided here, the NicheMapR package includes vignettes relevant to the ectotherm model that provide greater detail on model inputs and outputs (Supplementary material Appendix 1–2).

The NicheMapR ectotherm model structure

The ectotherm model is run with the R function `ectotherm` which receives the input parameters and microclimate forcing from the user. By default it looks for microclimate outputs in a list called 'micro' (this can be overridden by specifying the microclimate inputs, e.g. metout = metout_2). Thus the simplest operation with all default settings using the New et al. (2002) global monthly average climatology is:

```
> library(NicheMapR)
> micro <- micro_global()
> ecto <- ectotherm()
> environ <- as.data.frame(ecto$environ)
> plot(environ$TC ~ micro$dates)
```

The main computations are done by a Fortran library to which the `ectotherm` function connects via the wrapper function `ectorun`. The Fortran library of the NicheMapR ectotherm model, as described in this note, comprises 28 subroutines and seven functions (Fig. 3). There are additional functions and subroutines for solving transient heat budgets, simulating a container that fills with rainwater, and for the specific effects of butterfly wings, which will be described in future papers.

The subroutine `ECTOTHERM` (Fig. 3a) acts as the I/O to R via `ectorun` and controls each day's iteration. It receives the microclimate (two shade scenarios) and functional trait data provided from within R (Fig. 3b) and then sets up the beginning environment for a given hour's simulation as a function of the activity and thermoregulatory characteristics chosen (e.g. retreating below ground, climbing, seeking shade) (Fig. 3c). It then computes body temperatures through calls to `FUN` (Fig. 3d) and makes any necessary adjustments to find the optimal location in the habitat and the resultant body temperature and evaporative water loss. It will then optionally call the DEB models (Fig. 3g) and the `PHOTOPERIOD` function to determine the life cycle trajectory and finally store the results via `TRAPHR` and return them to the R environment (Fig. 3h). The `ECTOTHERM` subroutine also makes frequent calls to the `GEOM` subroutine which computes surface areas, volumes and linear dimensions given the type of organism chosen (plate, cylinder, ellipsoid, 'lizard', 'frog', or a customised option) and the changes imposed by any behaviours (e.g. postural changes).

The ectotherm model is designed to capture a variety of thermoregulatory responses of different kinds of organisms, only some of which may be relevant for a given taxon (e.g. burrowing, panting). A generic procedure for emulating behavioural thermoregulation is achieved through a cluster of six subroutines (Fig. 3c) that encode a fixed sequence of potential thermoregulatory behaviours (Fig. 4). The routines

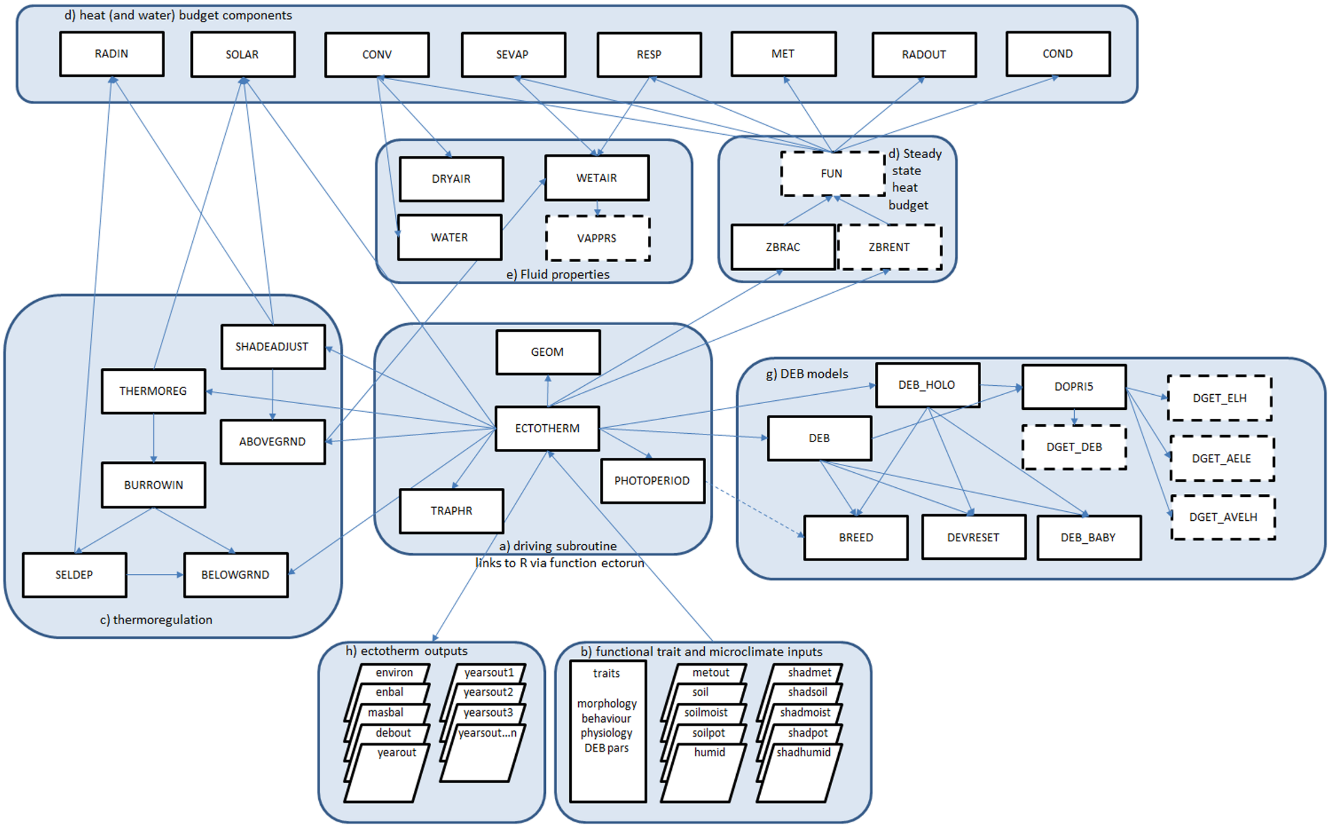

Figure 3. Structure of the Fortran library of the NicheMapR ectotherm model and its connection to the R environment. Solid boxes represent Fortran subroutines and dashed boxes represent Fortran functions. Arrows point to routines or functions called by a particular subroutine. The subroutine 'ectotherm' is the main driving program, receiving the input data, calling all other subroutines directly or indirectly, and returning the output. See main text for further explanation.

ABOVEGROUND and BELOWGROUND contribute to defining the microclimate the organism would experience if it were to be on the surface or in an underground retreat and are called as required depending on which microhabitat the algorithm is considering. Thermoregulation is spatially implicit with the options bounded by the extremes of shade dictated by the two sets of microclimatic data used as input. The subroutine SHADEAJUST alters 'shade' in user-specified increments and then calls SOLAR and RADIN to recompute the experienced radiative environment. After a change in shade, ABOVEGROUND adjusts the experienced solar radiation levels and air, sky and substrate temperature by linear interpolation between the shade extremes. Any ensuing changes in air temperature necessitate recalculation of the relative humidity, which is done through calls to the function WETAIR.

The organism may be simulated to choose any range of the ten available soil nodes as a retreat, and this is achieved via the subroutine BURROWIN which first chooses between shaded and unshaded underground conditions. It then calls the subroutine SELDEP which searches through the possible soil nodes until it finds the shallowest tolerable depth on the basis of critical thermal limits, given the current body temperature estimate (see below). SELDEP then calls BELOWGROUND and RADIN to define the new conditions. The underground environment is assumed to have nominally low wind speed and blackbody radiative conditions (i.e. uniform local soil and air temperature). The relative humidity experienced when underground is determined by the inputs of the soil humidity table from the microclimate model input or user defined microclimate input see (Kearney et al. 2018 for an example). The thermoregulatory process is more generally controlled by the subroutine THERMOREG, which assesses a fixed sequence of options for dealing with excessively hot or cold conditions given the body temperature thresholds and thermoregulatory options provided by the user as described in the 'Thermoregulation routines' section below.

The heat and mass (water exchange via evaporation) budget at steady-state is solved by subroutine FUN using a root finding method implemented via subroutine ZBRAC and function ZBRENT taken from Brent (2002). FUN receives inward fluxes of solar and infrared radiation indirectly via ECTOTHERM and calls MET, SEVAP, RESP, CONV, COND and RADOUT to compute the body temperature-dependent heat fluxes from metabolism, cutaneous evaporation, respiratory evaporation, convection, conduction and thermal long wavelength infrared loss, respectively (Fig. 3d).

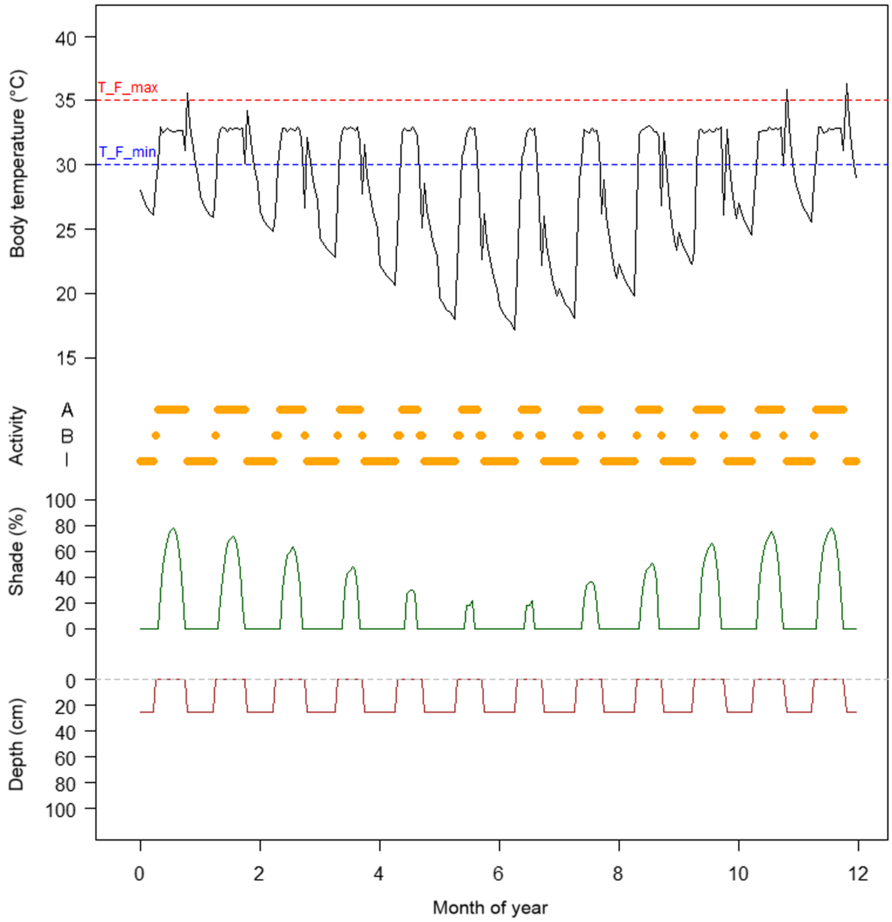

Figure 4. Daily cycles, for a typical day of each month of the year, of the body temperature (black), % shade selected (green, divided by 10 for plotting), activity level in orange (multiplied by 5 for plotting, so 5 = basking, 10 = active) and depth in brown (m) for the eastern water skink *Eulamprus quoyii* in Kuranda, Far North Queensland, Australia. Also superimposed are the foraging thresholds 'T_F_min' (blue dashed line) and 'T_F_max' (red dashed line), and the preferred or target temperature 'T_pref' (orange dashed line). Note how this lizard is active every day, but requires shade for this, and how body temperature jumps up on some evenings when it goes into its burrow (when inactive the body temperature may rise to a maximum of half way between 'T_F_max' and 'CT_max' before the animal moves deeper into the burrow).

The subroutine `CONV` adjusts air properties for temperature and pressure via subroutine `DRYAIR`, or water properties via `WATER` if an aquatic environment is simulated. Subroutines `SEVAP` and `RESP` call `WETAIR` to determine flows of heat and mass associated with evaporative heat exchange. `RESP` determines molar flows of respiratory gases given the oxygen consumption rate defined by the metabolic rate as computed in subroutine `MET`. The latter either uses an empirical, allometric function defined by the user to compute oxygen consumption from body mass and computed body temperature, or it uses the mechanistic mass and heat balance calculations of the DEB model directly if the DEB model is invoked.

Finally, the DEB model, when activated, is called once per hour by subroutine `ECTOTHERM` and is solved via the `DOPRI5` integration routines (Hairer et al. 1993) using relevant equations for the particular DEB model being invoked (see below for the types of DEB model) (Fig. 3g). The DEB model itself calls subroutine `BREED` to determine the seasonal pattern of reproduction, `DEVRESET` to determine how the life cycle resets within a simulation, and `DEB_BABY` if a viviparous organism is simulated to determine development time.

Pathways and parameters for heat and evaporative water exchange

The heat energy balance of an organism is the sum of the terms in the diagonal equation of Fig. 2. Each of these terms is computed by the ectotherm model each hour, with the resulting fluxes saved in the output table `enbal`.

The heat budget terms include specific animal parameters (functional traits) and forcing variables (functional environments or 'boundary conditions'). All the environmental terms require the surface area across which heat and mass is being exchanged. The ectotherm function has built-in calculations in subroutine GEOM for computing these areas for different shapes (e.g. shape = 1 is a cylinder) given relative proportions of linear dimensions provided as input (shape_a, shape_b, shape_c) as well as the density, rho_body, and wet weight, Ww_g, from which volume is computed. There are custom surface area/mass relations for the desert iguana and the leopard frog, and it is also possible to provide coefficients for custom functions representing the total area, ventral area, and the silhouette areas when the body is oriented normal and parallel to the sun's direct rays.

For infrared thermal radiation, QIRIN, and the incoming diffuse component of solar radiation, QSOL, the configuration factors (decimal fraction of the body surface looking) to the sky fatosk and ground fatosb must be specified. For infrared radiation lost, the emissivity of the outer surface, epsilon is required. For solar radiation, in addition to the silhouette areas and surface areas, the solar absorptivity of the substrate is required, alpha_sub, as well as the range of solar absorptivities of the animal itself (alpha_min, alpha_max).

For conduction, QCOND, the thermal conductivity of the substrate is required, k_sub, as well as the percentage of the surface area touching the ground, pct_cond. For convection QCONV, the shape- and size-specific heat transfer coefficient is computed within subroutine CONV using standard 'recipes' for determining dimensionless numbers (Nusselt, Reynolds, Grashof, Schmidt, Prandlt) for free and forced convection according to threshold Rayleigh (free) and Nusselt (forced) values.

The processes of evaporative water exchange, QEVAP, were described in detail by Tracy (1976) and more recently reviewed by Pirtle et al. (2019). As described in detail in Supplementary material Appendix D of Kearney and Porter (2004), the Colburn analogy between heat and mass transfer is used to determine the mass transfer coefficient for cutaneous water loss, and thereby both the heat and mass transfer of water from the skin. The functional traits driving this exchange are the percentage of the total surface area acting as a free water surface, pct_wet, and the percentage of the surface area taken up by the wet surface of the eye, pct_eyes and mouth pct_mouth Water loss from the eyes is only invoked when the animal is basking or active, because they are otherwise assumed to be closed. For respiratory water loss, the metabolic rate and respiratory quotient RQ together with the altitude-adjusted atmospheric pressure and environmental gas concentration determines the flow of air through the lungs, as mentioned earlier, as well as the oxygen extraction efficiency F_O2 and the temperature difference between inspired and expired air delta_air. Water loss from the mouth is only invoked when panting is occurring. The function get_p_wet of the NicheMapR package provides code for partitioning experimentally observed evaporative water loss into its different pathways and estimating many of these parameters. The output table masbal provides the mass flow of water due to these processes as well as the estimated oxygen consumption rate. It also produces many more mass fluxes when the DEB model is invoked, as described in detail in Supplementary material Appendix 2.

The ectotherm model also has scope to estimate transient heat budgets, and thus stored heat, Q_S, but without thermoregulatory behaviour. This functionality can also be used to estimate water temperatures in a 'bucket model' simulation of a container filling with rainwater which can be used to simulate ponds and other waterbodies under certain assumptions. These aspects of the model will be described elsewhere.

Thermoregulation routines

Thermoregulatory behaviour in the ectotherm model depends on the behavioural options relating to activity phase, microhabitats used, body temperature thresholds inducing particular behaviours, and thermal physiology (Supplementary material Appendix 1). The resulting environment experienced, and the associated body temperatures, are in the output table environ.

Organisms can be specified as any combination of nocturnal (nocturn = 1), crepuscular (crepus = 1) or diurnal (diurn = 1) in their activity phase and can be permitted to seek shade (shadeseek = 1), climb (climber = 1) or retreat underground (burrow = 1). The geographical coordinates, elevation and time of year dependent calculations of zenith angle and solar radiation are used to determine if the photophase is suitable for activity.

Minimum and maximum shade can be specified per day of the simulation (arrays minshades, maxshades) although the actual convective, evaporative and radiative environments available will be limited by the microclimate inputs. For example, one could simulate a microclimate with a maximum shade of 50% representing, say, bushes with sparse foliage. However, a sufficiently small animal could be allowed to seek 100% shade in a small patch (i.e. by setting maxshades to 100%) under the bush, thus being in a 50% shaded microsite with a concomitant air and substrate temperature, but not exposed to any direct solar radiation. Conversely an organism could be simulated to be in a shaded forest environment (high shade for the microclimate model run) but basking in a sun fleck (exposed to direct sun, i.e. by setting minshades to 0%).

If climbing is allowed, organisms can move to the reference height (typically ~ 1.5–2.0 m) from the ground and be exposed to a milder air temperature, different humidity and higher wind speed. Retreating below ground can be limited to a specified range of the simulated soil nodes (mindepth, maxdepth), as mentioned above, perhaps representing the nature of the terrain (e.g. shallow bedrock limiting the maximum depth) or the ability of the organism to retreat underground (e.g. too large to utilise existing rock crevices). An organism can also be simulated to choose shaded or unshaded underground refuges (shdburrow).

The activity state of the organism is either inactive, basking or foraging. The thermoregulatory sequence in the ectotherm model assumes that organisms are aiming to maximise activity at the preferred body temperature, T_pref, and will only be 'active' (i.e. moving beyond their retreat, catching prey, finding mates) between the minimum and maximum body temperature thresholds for foraging, T_F_min and T_F_max. The transition from retreat to basking is only allowed if body temperature exceeds the threshold T_RB_min – i.e. it has a minimum emergence temperature. This captures the fact that animals may not be seen active on unusually warm winter days if they are overwintering in a deep retreat site below their body temperature threshold for emergence. The transition from retreat to basking can also be conditional on whether a warming signal is detected at the depth it has retreated to (warmsig=1; if warmsig=0 then emergence will happen whenever T_RB_min is exceeded in the retreat and conditions are suitable for basking or foraging). The transition to the basking state is also conditional on whether it would be warmer to be outside rather than inside the retreat.

If conditions are suitable for emergence but too cool to bask or forage, a basking posture perpendicular to the sun's rays will be invoked. This maximises the silhouette area (computed in subroutine GEOM) exposed to the direct solar beam. The animal is simulated to also increase its area contacting the ground by 20% if the ground is warmer than it is.

Foraging is only permitted if T_F_min can be reached in a posture with an averaged silhouette area and a typical fractional area of contact between the body and the ground as determined by pct_cond. If conditions become warmer than T_pref the animal will first adjust its solar absorptivity to the minimum value alpha_min (in 5% intervals), then change its posture to be parallel to the sun's rays. If these colour and postural changes are insufficient, it will seek shade if permitted, up to the maximum specified value. Failing this, it will raise its preferred temperature until T_F_max and then climb (if permitted). Finally, if the parameter pantmax is set to greater than one, it will increase the airflow through the lungs by this multiplier at 0.1 increments with oral evaporation being invoked via the parameter pct_mouth which increases the area acting as a free water surface. If all these options fail, it will then seek shelter underground (if permitted).

When below ground, the subroutine SELDEP finds the shallowest depth from the permitted range that keeps the organism from going below its critical thermal minimum CT_min or above the midpoint between its T_F_max and CT_max.

The Dynamic Energy Budget model

DEB theory is a thermodynamically formalised framework for capturing the 'macro-metabolic' processes of feeding, assimilation, storage, mobilisation, maintenance, development, growth, reproduction and senescence of an organism across the whole life cycle (Supplementary material Appendix 2 for an overview). It was developed by Kooijman over ~30 yr (Kooijman 1986a, b, 1993, 2000, 2010). There is a family of models under DEB theory which capture qualitatively different metabolic schemes, including 'von Bertalanffy growth' and more complex patterns such as the growth trajectory of holometabolus insects (Llandres et al. 2015). DEB theory is challenging to learn but a number of papers provide accessible introductions (Nisbet et al. 2000, Meer 2006, Jusup et al. 2017, Marques et al. 2018).

Earlier versions of the 'Niche Mapper' biophysical modelling framework (Porter et al. 1973, Kearney and Porter 2004) incorporated 'static energy budgets', i.e. considering only one particular life stage at a snapshot in time, using descriptive empirical functions of metabolic processes such as allometric functions for metabolic rate. DEB theory provides a more mechanistic basis for doing this dynamically through the ontogeny, thereby capturing the temporal dimension of the niche.

There are natural feedbacks between the biophysical and DEB models, including the interaction between size, shape, heat and water exchange, the interaction between feeding and activity, and the interaction between metabolism and the water budget. The integration of biophysical models with DEB theory thereby provides a holistic picture of the heat, water, energy and nutritional budget and the associated life history consequences in a given sequence of environments.

Currently the DEB models that can be implemented in the NicheMapR ectotherm model are 'standard' (von Bertalanffy type growth under constant food, many birds and reptiles), 'abj' (early growth acceleration, many fish), 'abp' (acceleration to maturation and cessation of growth – many hemimetabolous insects) and 'hex' (holometabolus insect). The 'DEB model tutorial' vignette, Supplementary material Appendix 2, provides a brief introduction to the conceptual framework of DEB theory and describes the input parameters, how to obtain them from the Add My Pet database (Marques et al. 2018, > 2000 species at the time of writing) and how to use the R DEB functions in NicheMapR as well as the ectotherm model in conjunction with the DEB model. In these appendices and in the examples below, the potential of the DEB approach is illustrated in terms of the kinds of outputs it can generate (for examples of previous studies using the DEB module of NicheMapR, see Kearney 2012, 2013, Kearney et al. 2013, Schwarzkopf et al. 2016).

In addition to the standard DEB parameter inputs (Supplementary material Appendix 2), the NicheMapR implementation of the DEB models requires parameters to define the timing of reproduction which can be made a function of photoperiod, either for specific day lengths or in relation to the solstices and equinoxes (parameters photostart, photofinish, daylengthstart, daylengthfinish, photodirs, photodirf). It also requires extra parameters for the water budget, including the percentage of water in the food, faeces, urine (parameters pct_H_X, pct_H_P, pct_H_N, respectively) as well as the percentage of body water that can be lost before the animal enters aestivation `pct_H_R` (broken by a threshold rain event, `raindrink`) or dies `pct_H_death`.

A major advantage of the DEB model is that it naturally handles variable food levels. If food availability is known, this can be included as well on the basis of a Holling's type II functional response (Holling 1959) relating feeding rate to food density `X` via a half saturation constant `K`. Behavioural, thermal, nutritional and hydric parameters can be specified separately for each life stage (e.g. egg, juvenile, adult, instars, pupae) via the input matrices `behave_stages`, `thermal_stages`, `water_stages`, `nutri_stages` and `arrhenius`.

When the DEB model is invoked, the output table `masbal` includes values for the complete mass budget, including gas and water fluxes, food intake and faecal output. The output table, `debout`, is also produced which provides hourly output of the DEB state variables as well as the growth in length and mass, reproduction and aging, and the full partitioning of the biomass. Finally, the tables `yearout` and `yearsout` provide lifetime and annual summaries of life history information as well as indications of the timing and nature of significant events including death (through heat, cold, desiccation, starvation and senescence) as well as estimates of the finite rate of population increase R_0 and the intrinsic rate of population increase r_i. The latter are made on the basis of the reproduction trajectory together with hourly mortality rate estimates specific to periods of activity `m_a` and inactivity `m_i`, as well as in the first year since 'hatching' `m_h` (for examples, see Kearney 2012, 2013). A comprehensive application and test of this whole modelling system can be found in Kearney et al. (2018).

Example applications

To illustrate the basic use of the NicheMapR ectotherm model, we first simulate the body temperature and thermoregulatory behaviour of a lizard, the Australian eastern water skink *Eulamprus quoyii*, for the middle day of each month of the year. We then illustrate a dynamic energy budget model for this species under constant conditions using the R DEB functions of the NicheMapR package. Finally, we show an example of the DEB model running via the ectotherm model, predicting the growth and reproduction trajectory under simulated natural microclimatic conditions. Further details, code and additional examples are provided in the vignettes 'Ectotherm Model Tutorial' (Supplementary material Appendix 1) and 'DEB Model Tutorial' (Supplementary material Appendix 2) included in the package.

```
First, install the package.
> install.packages('devtools')
> library(devtools) # load the devtools package
> install_github('mrke/NicheMapR')
> library(NicheMapR) # load the NicheMapR package
> get.global.climate() # this will download and unzip a 0.5GB global monthly climate data
```

Instructions for package installation were also provided in Kearney and Porter (2017). However, in contrast to the situation at the time of the latter publication, all Fortran source code is now included with the package and can thus be compiled locally, producing a shared object 'NicheMapR.dll' (Windows) or 'NicheMapR.so' (OSX or Linux), which contains the microclimate, ectotherm and global aerosol database subprograms. A Fortran compiler must be available for R. Rtools <https://cran.r-project.org/bin/windows/Rtools/> will achieve this for Windows, and OSX gfortran must be installed, e.g. via <https://github.com/fxcoudert/gfortran-for-macOS/releases>.

Next, run the microclimate model for Townsville, Australia, using the global climate database and all the default settings, and then run the ectotherm model with all default settings.

```
> library(NicheMapR)
> longlat <- c(146.77,-19.29) # Townsville, northern Australia
> micro <- micro_global(loc=longlat) # run the microclimate model
> metout <- as.data.frame(micro$metout) # retrieve above ground microclimate
> ecto <- ectotherm() # uses default settings (the Eastern Water Skink)
```

The next code example will extract the results table 'environ' and plot some of the outputs contained within, to show the pattern of body temperature, activity and thermoregulation over the middle day of each month (Fig. 4).

```
> # retrieve output
> environ <- as.data.frame(ecto$environ) # behaviour, Tb and environment
> environ <- cbind(environ, metout$SOLR) # add solar radiation for activity window plots
> colnames(environ)[ncol(environ)] <- "Solar"
> # append dates
> dates <- micro$dates
> metout <- cbind(dates, metout)
> environ <- cbind(dates, environ)
> # Plot the body temperature, shade, activity and depth selected through time.
> # Hourly Tb (black), activity (orange, I=inactive, B=basking, A=foraging), shade (green) and depth (brown)
> with(environ, plot(TC ~ dates, ylab="", xlab="month of year", col=‚black', xlim=c(-0.25, 12), ylim=c(-20, 40), type="l", yaxt='n'))
```

```
> with(environ, points(ACT * 2+7 ~ dates,
  type="p", pch=16, col="orange"))
> with(environ, points(SHADE / 10 - 6 ~
  dates, type="l", col="dark green"))
> with(environ, points(DEP - 10 ~ dates,
  type="l", col="brown"))
> abline(ecto$T_F_min, 0, lty=2, col=
  'blue')
> abline(ecto$T_F_max, 0, lty=2, col=
  'red')
> ytick<-seq(15, 40, by=5)
> axis(side=2, at=ytick, labels=TRUE)
> mtext(text=c('A', 'B', 'I'), side=2, line=1,
  at=c(11, 9, 7))
> ytick<-seq(-6, 4, by=2)
> axis(side=2, at=ytick, labels=FALSE)
> mtext(text=seq(0, 100, 20), side=2,
  line=1, at=seq(-6, 4, 2), las=2)
> ytick<-seq(-20, -10, by=2)
> axis(side=2, at=ytick, labels=FALSE)
> mtext(text=rev(seq(0,100,20)),side=2,
  line=1, at=seq(-20, -10, 2), las=2)
> abline(h=-10, lty=2, col='grey')
> mtext(text=c('body temperature (°C)',
  'activity', 'shade (%)', 'depth (cm)'),
  side=2, line=2.5, at=c(30, 9, 0, -15))
> text(-0.2, c(ecto$T_F_max+1, ecto$T_F_
  min+1), 'T_F_max', col=c('red', 'blue'),
  cex=0.75)
```

Another useful way to look at the activity window is shown below, producing Fig. 5. This kind of plot was first done for the Desert Iguana in Porter et al. (1973).

```
> # seasonal activity plot (dark blue=
  night, light blue=basking, orange=
  foraging)
> forage <- subset(environ, ACT == 2) #
  get foraging hours
> bask <- subset(environ, ACT == 1) # get
  basking hours
> night <- subset(environ, Solar == 0) #
  get night hours
> with(night, plot(TIME ~ DOY, ylab="Hour
  of Day", xlab="Day of Year", pch=15,
  cex=2, col='dark blue')) # nighttime
  hours
> with(forage, points(TIME ~ DOY, pch=15,
  cex=2, col='orange')) # foraging Tbs
> with(bask, points(TIME ~ DOY, pch=15,
  cex=2, col='light blue')) # basking Tbs
```

To run the DEB model, parameters for a species in the AmP collection can be extracted from the allstat.mat file with the following code:

```
> install.packages('R.matlab')
> library(R.matlab)
> allStat <- readMat('allStat.mat') # this
  will take a few minutes
> save(allStat, file='allstat.Rda') # save
  it as an R data file for faster future
  loading
```

Then a species can be chosen, and a simulation of the trajectory of growth and reproduction computed and plotted for a given temperature and food level (Fig. 6a) as well as the corresponding mass fluxes (Fig. 7).

```
> species <- "Eulamprus.quoyii" # must
  be in the AmP collection - see allDEB.
  species list
> ndays <- 365*5 # number days to run
  the simulation for
```

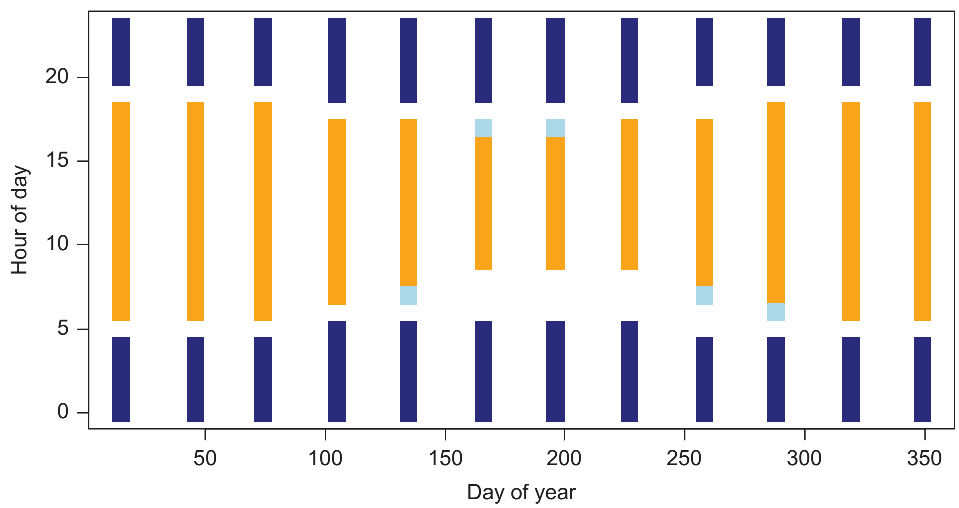

Figure 5. The annual activity window for the eastern water skink *Eulamprus quoyii* in Kuranda, Far North Queensland, Australia on the basis of Fig. 4, with dark blue representing the nighttime hours, light blue the basking hours and orange the foraging hours. Activity is possible for this lizard at this tropical location throughout the year.

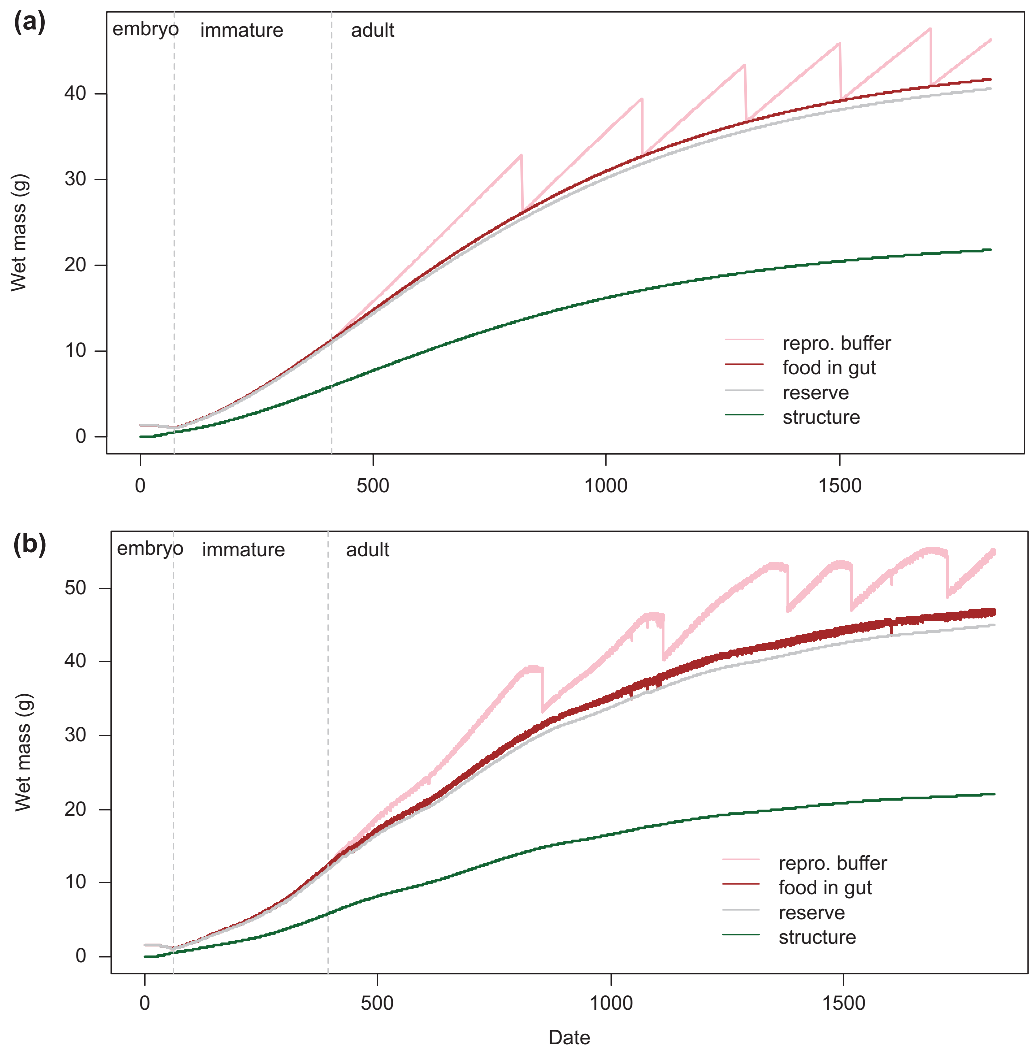

Figure 6. Example Dynamic Energy Budget growth trajectory of the Australian eastern water skink *Eulamprus quoyii*, generated by (a) the R DEB function of NicheMapR and (b) the ectotherm model of NicheMapR. The former is under constant food and temperature, with no activity constraints or seasonal reproduction. The latter shows the lizard growing at Townsville Australia (Schwarzkopf et al. 2016) under a realistic microclimate. These plots and the underlying code are explained in detail in the Supplementary material Appendix 2 'Introduction to the DEB model'.

```
> div <- 1 # time step divider (1=days,
  24=hours, etc.) - keep small if using
  Euler method
> Tbs <- rep(25, ndays * div) # °C, body
  temperature
> X <- 100 # J/cm2 base food density
> Xs <- rep(X, ndays) # food density (J/cm2
  or J/cm3)
> E_sm <- 350 # J/cm3, volume-specific
  stomach energy
> clutchsize <- 5 # -, clutch size
> mass.unit <- 'g'
> length.unit <- 'mm'
> plot <- 1 # plot results?
> start.stage <- 0 # stage in life cycle
  to start (0=egg, 1=juvenile, 2=puberty)
> deb <- rundeb(species=species, ndays=
  ndays, div=div, Tbs=Tbs, clutchsize=
  clutchsize, Xs = Xs, mass.unit=mass.
  unit, length.unit=length.unit, start.
  stage=start.stage,E_sm=E_sm,plot=plot)
```

In Supplementary material Appendix 2, code is provided to run the ectotherm model for *Eulamprus quoyii* with the DEB model turned on at the Townsville site, to produce a growth and reproduction trajectory that accounts for variable body temperature and foraging periods (Fig. 6b).

Discussion

The capacity to compute body temperatures, metabolic rates, activity times and evaporative water losses under realistic

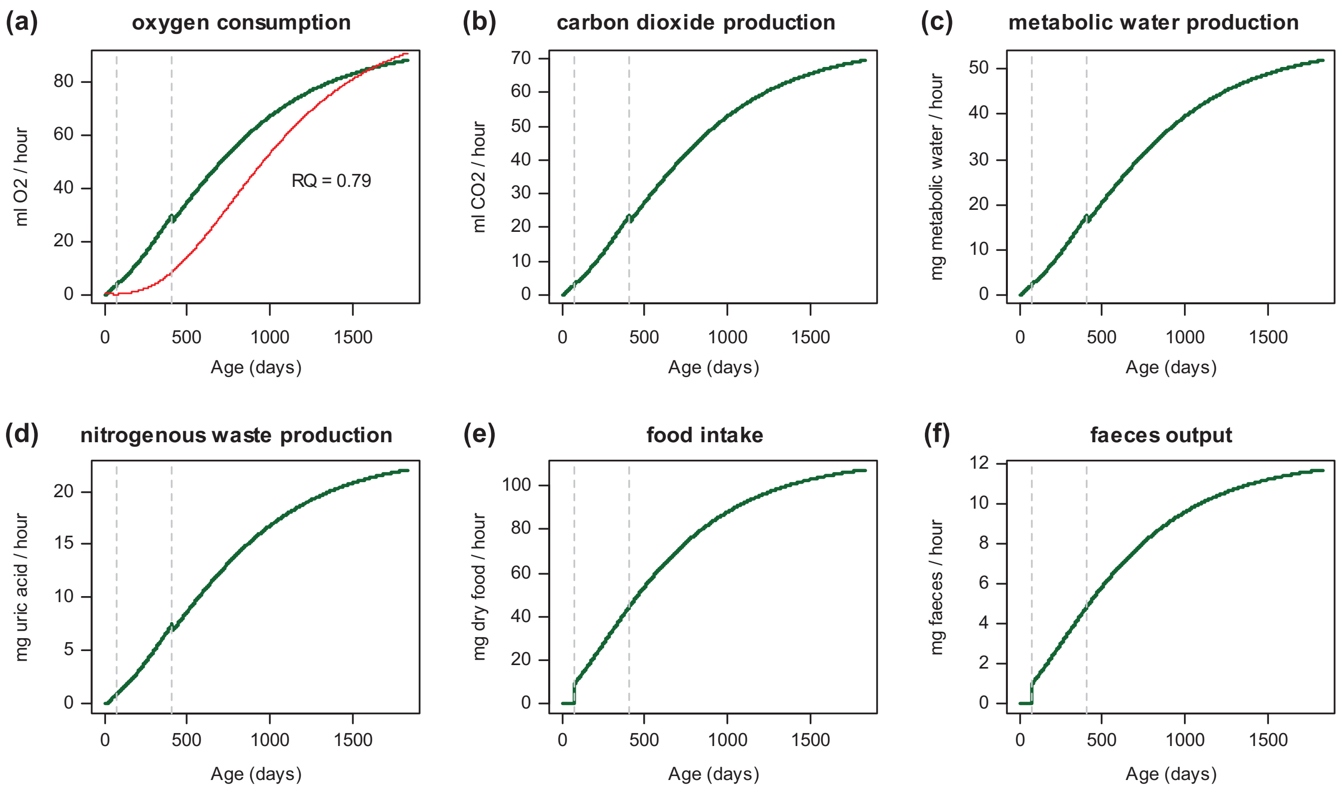

Figure 7. Dynamic Energy Budget (DEB) model predictions of some mass fluxes for the lizard *Eulamprus quoyii* growing under constant ad libitum food at 25°C. In panel (a), the red line is the prediction from a general allometric relation for adult lizards (Andrews and Pough 1985). 'RQ' is the respiratory quotient, the ratio of CO_2 O_2 production. No data of this type was used to fit the DEB parameters for this species but these responses emerge naturally from the stoichiometric framework of the DEB theory.

microclimates allows many inferences to be made about the ability of different types of animals to live in different climates and habitats. In some cases, such processes may be limiting factors of the distribution. For example, Kearney and Porter (2004) found that activity time was likely to be a limiting factor for a nocturnal lizard in southern Australia. More often, these constraints act to limit the overall energy and water budget in a subtler manner through their effects on development, growth and reproduction. These effects can be captured through the DEB model, which allows the temporal dimension of the niche to be included explicitly (Kearney et al. 2018). Because the DEB model predicts reproductive success and survival, it can inform the vital rates of population models thereby transcending the individual level (Fig. 1).

The mechanistic niche models that can be developed with NicheMapR are parameter rich, because they are process rich. Many of these parameters can be well defined empirically, and often represent routinely-measured eco-physiological traits. Per process, there are in fact very few parameters and this is in part because of the advantages conferred by the framing of the models with well-established physical and chemical principles. Not only does this thermodynamic framing avoid the extrapolation problem that would arise if descriptive models of physiological processes were used in their place, but it also means that relatively simple models can predict complex dynamics under highly variable and potentially novel environmental sequences. The NicheMapR package should therefore be useful for understanding and predicting how species respond to climate, habitat and resource changes under past, present and future conditions.

Online resources

The NicheMapR R package source is currently available at <https://github.com/mrke/NicheMapR>, with the latest release at <https://github.com/mrke/NicheMapR/releases> and a website with general information at <https://mrke.github.io/>. For help and discussion, join the Google Group <https://groups.google.com/forum/#!forum/nichemapr>. For DEB model parameters, see <www.bio.vu.nl/thb/deb/deblab/add_my_pet/> and <https://github.com/add-my-pet/AmPtool>, and for DEB parameter estimation tools see <https://github.com/add-my-pet/DEBtool_M> and <https://github.com/add-my-pet/DEBtool_R>. The Zenodo doi for the package is: <https://zenodo.org/badge/latestdoi/45649715>.

To cite NicheMapR or acknowledge its use, cite this Software note as follows, substituting the version of the application that you used for 'version 0':

Kearny, M. R. and Porter, W. P. 2019. NicheMapR – an R package for biophysical modelling: the ectotherm and Dynamic Energy Budget models. – Ecography 42: 000–000 (ver. 0).

Acknowledgements – We thank Lauren Buckley and two anonymous reviewers for helpful feedback on the MS.
Funding – This research was supported by an Australian Research Fellowship (MRK) from the Australian Research Council, DP110102813.
Author contributions – MRK designed and wrote the R software, Dynamic Energy Budget algorithms and documentation, WPP and MRK designed and wrote the Fortran software, MRK wrote the manuscript with contributions from WPP.

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Supplementary material (Appendix ECOG-04680 at <www.ecography.org/appendix/ecog-04680>). Appendix 1–2.